Gas Turbine Simplified Digital Twin

Probabilistic Performance & Cashflow Framework for LTSA-Constrained Combined-Cycle Assets

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The Investor Challenge: Gas Turbine Performance Risk

$22–30M
Major inspection cost per event (2-unit CC block)

0.8–1.5%
Annual heat rate degradation eroding fuel margin daily

350 EOH
Single cold start penalty — equivalent to 15 days running

2–5×
Creep-fatigue interaction multiplier vs. independent damage

Key Drivers of Performance & Cost Uncertainty

Operating Profile Uncertainty
Cycling-intensive dispatch accumulates EOH 2–3× faster than baseload. Start type mix (cold/warm/hot) and load swings drive both contractual maintenance timing and physical fatigue damage.

Climate & Market Correlation
Ambient temperature drives capacity derating and heat rate simultaneously. Summer peaks concentrate high-price dispatch at worst thermal efficiency — amplifying fuel cost and stress accumulation.

LTSA Coverage Gaps
CSA contracts cover scheduled inspections but exclude generator, controls, BOP, and many unplanned failures. The gap between covered and actual exposure can reach $2–5M per event.

Damage Mechanism Interaction
Creep and fatigue interact synergistically — components fail earlier than either mechanism alone predicts. Standard EOH counting ignores this interaction, underestimating true risk.

The Modeling Gap: Why Standard Approaches Fall Short

Traditional LTSA Performance Assessment

— Relies on OEM's EOH count as sole life metric

— Treats creep and fatigue as independent — misses interaction damage

— Linear compressor fouling assumptions with optimistic wash recovery (80–90%)

— Static forced outage rate from historical fleet averages

— Deterministic single-scenario projections — no probabilistic distribution

— Ignores HRSG/ST cycling damage in start cost allocation

— No dispatch mode sensitivity — assumes fixed operating strategy

InfraSure Simplified Digital Twin

— Coupled creep-fatigue interaction model (ASME N-47 envelope)

— Non-linear compressor fouling with site-class coefficients (60–80% recovery)

— Weibull TBC failure model — path-specific sampling captures early failure tail

— Endogenous forced outage prediction from engineering stress state

— 1,000+ correlated climate × price simulations → P10/P50/P90 distributions

— HRSG drum fatigue tracker + split GT/HRSG start costs for 1×1 modeling

— Three dispatch modes test operator strategy trade-offs across the ensemble

Why This Framework Works for Investor Due Diligence

A purpose-built probabilistic model designed for the precision, scalability, and defensibility required in infrastructure investment analysis.

No OEM Data Required

Uses published GER-3620K maintenance guidelines, EPRI fleet statistics (1026609, 1025357), ASME N-47 interaction methodology, and NREL cycling cost data. Every model component traces to peer-reviewed or publicly available engineering references — defensible under independent engineer review.

Forward-Looking Climate Integration

Climate simulations with temperature autocorrelation and heat-wave persistence drive ambient conditions — not historical averages. Capacity derating and heat rate degradation respond to the climate the asset will actually face, capturing the correlation between high-price dispatch and worst thermal efficiency.

Endogenous Risk Prediction

Forced outage events emerge from the engineering stress state — combustion fatigue index, TBC Weibull threshold, rotor life fraction — rather than being imposed as static fleet-average assumptions. The model generates risk, it doesn't assume it.

Dispatch Strategy Sensitivity

Three operating modes (maximize dispatch, balanced, minimize LTSA cost) run the same climate and price scenarios with different EOH penalty structures. The investor sees the shape of the trade-off: how much revenue is sacrificed to improve tail risk, and whether conservative dispatch is clearly dominant.

Framework Architecture: 5-Step Modeling Pipeline

An integrated pipeline from climate simulation to investor cashflow metrics — each step builds on the last.

1

Climate Simulation

Temperature & weather paths

Correlated daily ambient temperature series with autocorrelation and heat-wave persistence for NYISO Zone F.

PRE-BUILT

2

Dispatch Model

Commitment & economic dispatch

Hourly commit/decommit optimisation against spark spread with mode-dependent EOH proximity penalties.

PRE-BUILT

3

Engineering Model

Simplified digital twin

8 stress accumulators with coupled creep-fatigue interaction, non-linear fouling, Weibull TBC, HRSG cycling.

THIS FRAMEWORK

4

Maintenance & Failure

Inspections & forced outages

EOH threshold scheduling, endogenous forced outage from stress state, LTSA coverage classification.

THIS FRAMEWORK

5

Financial Metrics

P10/P50/P90 distributions

EBITDA, DSCR, capacity factor, insurance adequacy across the full simulation ensemble.

PRE-BUILT

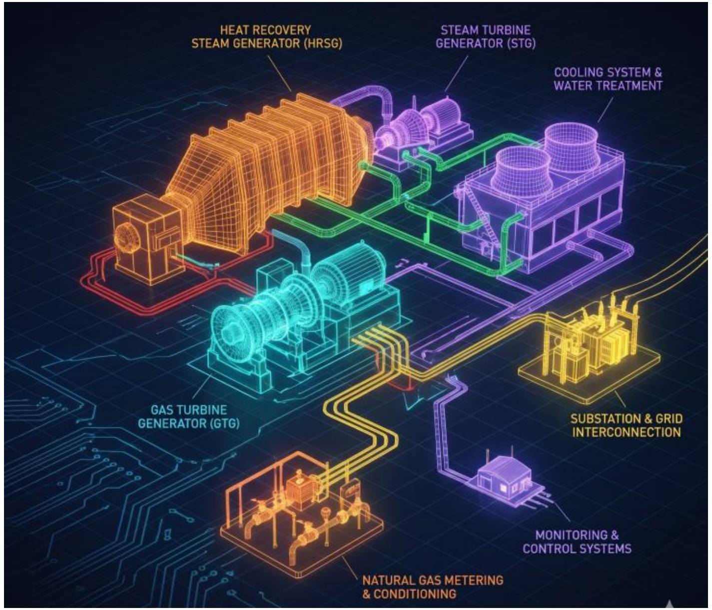
Daily feedback: degraded HR & capacity update next day's dispatch economics

Engineering Model: From Dispatch to Damage

From Daily Operating Profile to Component Life Consumption

Dispatch Profile → Stress Accumulation → Damage Interaction → Threshold Check → Outage or Continue

Fired Hours & Starts	Daily dispatch produces fired hours, start type (hot/warm/cold/trip), load profile, and HRSG restart classification. These raw operating metrics are the input to all stress calculations.
EOH + Creep-Fatigue Coupling	Contractual EOH tracks inspection timing. Parallel physics-based model evaluates Robinson life-fraction (creep) + Miner's rule (fatigue) against ASME N-47 interaction envelope — components under combined loading fail 20–40% earlier.
Non-Linear Compressor Fouling	Exponential accumulation model: fouling_loss = A × (1 - e^(-t/τ)). Site-class coefficients (clean A=1.5%, humid coastal A=2.5%). Offline wash recovers 60–80%. The biggest single contributor to heat rate gap between CSA guarantee and actual.
Weibull TBC & Endogenous FOR	Coating failure sampled from Weibull (β≈3.0, η≈28K hrs) per simulation path. Daily forced outage probability driven by combustion fatigue, TBC state, rotor life, and HRSG condition — risk is predicted, not assumed.



Operating Stress Factors: What We Track and Why

Stress Factor	Priority	Primary Causes	Source	Impact	Methodology
EOH + creep-fatigue coupling	HIGH	Fired hrs, start type, fuel, trips, load swings	Dispatch + LTSA	Maint. cost timing, planned outages	GER-3620 EOH + ASME N-47 envelope
Capacity derating	HIGH	Ambient temp, humidity, altitude	Climate + asset specs	Revenue loss, capacity shortfall	OEM correction curves -0.5%/°F above ISO
Heat rate degradation	HIGH	Fired hours, ambient, compressor condition	Dispatch + climate	Fuel cost +$400-600K/yr per 1%	Sawtooth 0.8-1.5%/yr partial resets
Combustion cycling fatigue	HIGH	Starts, trips, load swing magnitude	Dispatch	Forced outage, uncovered repair	Damage index cycling coupled via N-47
HRSG cycling damage	HIGH	Start type, temp differential	Dispatch	CC start cost (40-60% of total)	NREL drum fatigue split GT/HRSG costs
Compressor degradation	MED	Operating hours, air quality, wash sched.	Dispatch + asset specs	HR degradation (~50% of total)	Non-linear exponential 60-80% wash recovery
TBC coating life	MED	Firing temp, cycling, fuel contaminants	Dispatch + asset specs	Forced outage, blade replacement	Weibull (β≈3, η≈28K) path-specific sampling
Rotor life	LOW	Start-stop cycles, thermal transients	Dispatch	Catastrophic risk, 6-12 mo lead time	Life fraction vs. design life (5-10K starts)

Daily Time-Stepping Engine: Simplified Flowchart

Each simulated day: read state → apply degradation → compute economics → check for forced outage → dispatch decision → update stress → loop.

Read state vector +
day-d climate & prices

Apply degradation →
eff. capacity, HR, gas cost

Compute marginal cost
fuel + V-O&M + start (GT + HRSG)

Forced outage triggered
duration + cost + classification

Forced outage check
P_forced(stress state)

Dynamic start cost adjustment
As EOH approaches LTSA threshold, wear cost multiplier escalates (1.0x → 2.5–4.0x), penalizing marginal dispatches and discretionary cold starts.

Dispatch decision (mode-dependent)
margin vs. dynamic EOH proximity penalty

in the money

Operate
record hours, starts, load

out of the money

Idle
increment shutdown hours

Update stress accumulators
EOH + D_c + D_f (coupled) | fouling |
TBC | HRSG drums | rotor life

Check thresholds
EOH ≥ inspection?
Interaction envelope?

Write state → d+1
Loop until projection
horizon exhausted

Dispatch Modes: Testing Operator Strategy Trade-Offs

Understanding differences in dispatch strategy, EOH accumulation, and financial outcome is critical for investment thesis formation.

Dimension	Mode A: Maximize Dispatch	Mode B: Balanced	Mode C: Minimize LTSA Cost
EOH proximity penalty	None (1.0×)	Non-linear: 1.0× far, 2.5× within 1,000 EOH	Steep: 1.0× far, 4.0× within 2,000 EOH
Start cost adjustment	Base costs only	Wear component scales with proximity	Wear scales aggressively; cold starts heavily penalized
Dispatch behavior	Dispatches whenever margin > 0	Free when far from threshold; self-curtails on marginal days	Only high-margin days near thresholds; maximizes interval
Revenue impact	Highest gross revenue	Moderate sacrifice	Lowest gross revenue
Maintenance timing	Earliest inspections	Moderately deferred	Latest inspections
Forced outage exposure	Highest risk	Moderate risk	Lowest risk
Trade-off	*Maximum revenue, maximum risk*	*Revenue-risk balance, most realistic*	*Minimum risk, significant revenue loss*

Validation: Literature Review & Forward Plan

Framework validated against peer-reviewed research and industry practice. All identified gaps have been incorporated into v1.1.

Literature Issues Found & Fixed

Creep-fatigue interaction → ASME N-47 coupled envelope
Compressor fouling → non-linear model, 60-80% recovery
HRSG damage → drum tracker + split start costs
Forced outage → endogenous from stress state
TBC life → Weibull failure model ($\beta \approx 3.0$)
Part-load HR → fitted quadratic polynomial

Pilot Implementation

Athens-type GE 7FA 2×1 CC in NYISO Zone F
22-year mid-life asset, just completed HGP at 24K EOH
TGP Zone 6 gas supply with firm lateral transport
Full LTSA template with inspection schedule through MI at 48K
Initial state vector: 18 variables across GT, HRSG, and BOP

Sensitivity Analysis Plan

Creep-fatigue envelope shape: D_{int} = 0.6 vs 0.8
Compressor fouling site class: clean vs. humid vs. industrial
Weibull TBC β parameter: 2.5 vs. 3.0 vs. 4.0
Damage index calibration for partial load swings
Dispatch mode convergence: do Modes A–C produce meaningfully different EBITDA and DSCR distributions?

Cross-Validation & Expert Review

Compare model-predicted FOR against NERC GADS 7FA fleet data
Validate inspection timing against historical plant records
Independent engineer review of creep-fatigue parameterisation and simplified cycle counting calibration
Borescope / inspection data comparison where available

Built on Validated Research

References & Data Sources

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